

Backend Architecture & Auth Implementation

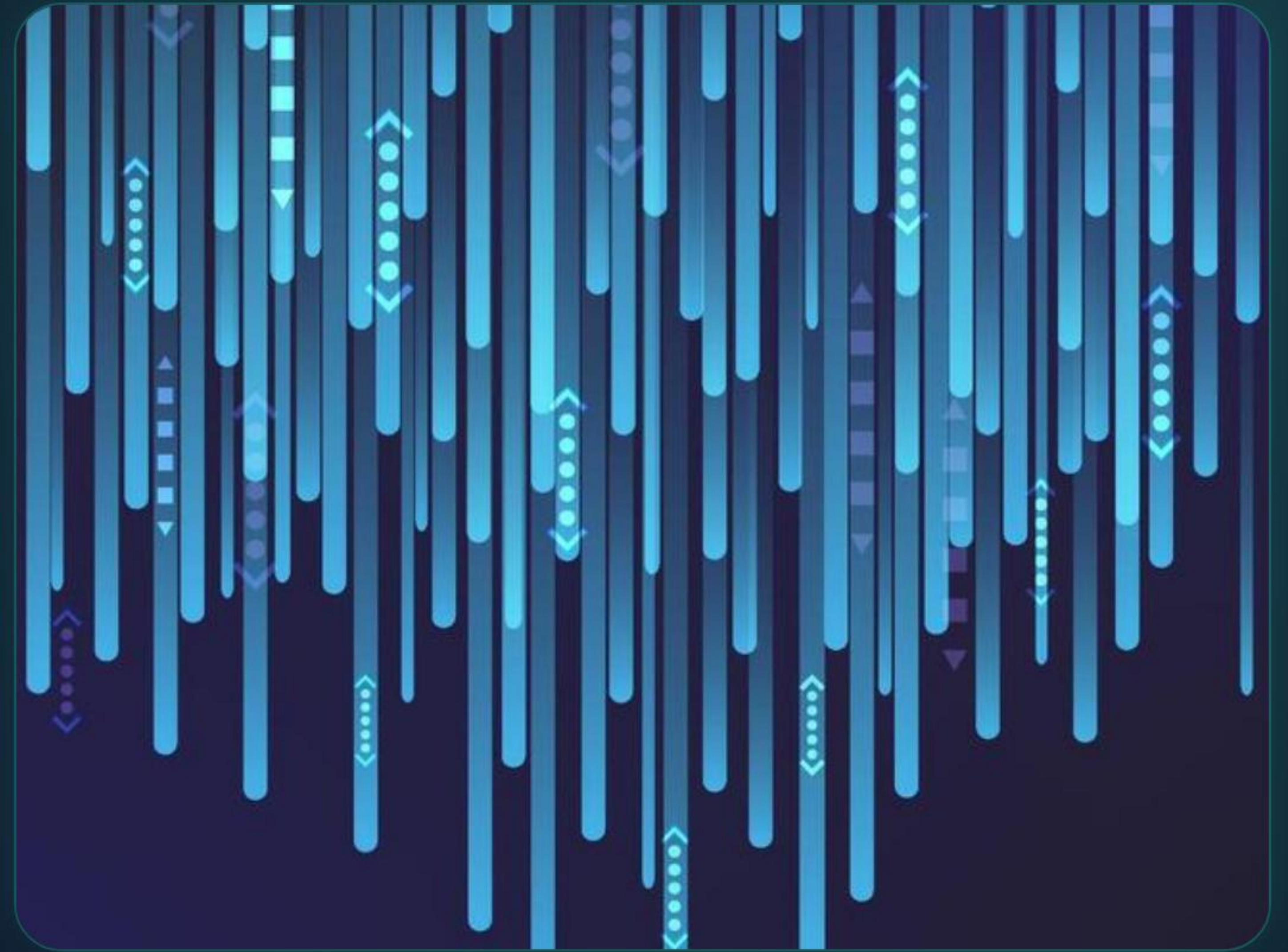
Receipt & Invoice Digitizer

MOHAMMAD MASTAN VALI SHAIK

Backend Development & Integration

PROJECT OVERVIEW

-  **User Authentication:** Robust handling of secure login and registration.
-  **Secure Processing:** Efficient data management and protection.
-  **API Communication:** Facilitating seamless internal data transfers.
-  **Frontend Integration:** Bridging the UI with backend logic.



| MY CONTRIBUTION

- ✓ Designed core **Flask backend** architecture.
- ✓ Developed secure **Login/Register** endpoints.
- ✓ Integrated backend logic with **HTML forms**.
- ✓ Implemented **bcrypt** password hashing.
- ✓ Configured and optimized **SQLite** storage.
- ✓ Resolved complex **integration bugs**.

| TECHNOLOGY STACK



Python & Flask

Core development framework for building scalable RESTful services.



SQLite

Lightweight relational database for structured user information storage.



bcrypt

Industry-standard password hashing to ensure cryptographic security.

| AUTHENTICATION FLOW



DATABASE & SECURITY

Structured Storage: Optimized schema for user profiles.

Encryption: Multi-layer hashing for data at rest.

Validation: Backend sanitization of all inputs.

Safety: Mitigation of common injection threats.

"Securing user identity is the core of system integrity."



PROJECT IMPACT



Core System Foundation

Established the critical backend structure required for the entire application life cycle.



Enhanced User Security

Implemented enterprise-level authentication to protect sensitive user digitizations.



Seamless Integration

Successfully unified backend services with frontend elements for a fluid user experience.

Summary of Contributions

MOHAMMAD MASTAN VALI SHAIK

Backend Development Lead

Receipt & Invoice Digitizer Project

Secure

Scalable

Integrated

IMAGE SOURCES



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